COURSE-BASED PROJECT DOCUMENTATION | DOC-04

Grammar Source & Bison Adaptation

ANTLR4 Reference Grammar Adaptation to LALR(1) Shift-Reduce

1. Grammar Source and Origin

The reference grammar for this project was derived from the official ANTLR4 Java grammar repository: grammar/original/JavaLexer.g4 and grammar/original/JavaParser.g4. ANTLR4 uses an ALL(*) top-down recursive-descent parsing algorithm with arbitrary lookahead and semantic predicates. Because GNU Bison uses a deterministic bottom-up LALR(1) algorithm, the reference grammar had to be structurally adapted.

2. Fundamental Adaptations for GNU Bison LALR(1)

A. Left-Recursion Conversion

ANTLR4 expresses repetitive lists using EBNF operators (*, +). In Bison LALR(1), lists are structured using left-recursive productions to ensure constant stack space during reduction:

```
// Left-recursive list in Bison LALR(1)
typeList:
      typeType
    | typeList COMMA typeType
    ;
```

B. Dangling-Else Disambiguation

Resolved using operator precedence directives to bind ’else’ to the innermost ’if’:

```
%nonassoc LOWER_THAN_ELSE
%nonassoc ELSE

statement:
      IF LPAREN expression RPAREN statement %prec LOWER_THAN_ELSE
    | IF LPAREN expression RPAREN statement ELSE statement
    ;
```

C. Java 21 Switch Guard Isolation (when clauses)

Disambiguated boolean guard expressions from switch case arrows (->) using a dedicated guardExpression nonterminal.

D. Ragged & Multidimensional Array Dimensions

Stratified array creation rules into explicit non-empty dimension expressions followed by optional empty brackets (new int[5][]).